

DOCUMENT APPROVAL SYSTEM

Tables and rules

1. Actors

Actor	Description	Primary Responsibilities
Employee	Document requester	Create requests, upload documents, respond to queries
Reviewer	First level reviewer	Review, comment, request additional information
Approver	Decision authority	Approve, reject, escalate
Senior Approver	Final authority	Final approval, override decisions
Compliance Officer	Audit stakeholder	Review audit trails and reports
System Administrator	System owner	Manage users, workflows and configurations

2. Primary Use Cases

Use Case ID	Use Case	Summary
UC-001	Create Document	Employee creates document request
UC-002	Submit Document	Submit draft into workflow
UC-003	Review Document	Reviewer reviews request
UC-004	Request Additional Info	Reviewer requests clarification

UC-005	Approve Document	Approver approves stage
UC-006	Reject Document	Approver rejects request
UC-007	Audit Document	Compliance officer reviews history

3. State Definitions

State	Meaning	Owner
DRAFT	Created but not submitted	Employee
SUBMITTED	Submitted to workflow	System
UNDER_REVIEW	Being reviewed	Reviewer/Approver
ADDITIONAL_INFO_REQUESTED	Waiting for clarification	Employee
RESUBMITTED	Clarification submitted	Employee
APPROVED	Workflow completed successfully	System
REJECTED	Workflow terminated	System
ARCHIVED	Closed and retained	System

4. State Transition Rules

Current State	Action	Next State	Allowed Role
DRAFT	Submit	SUBMITTED	Employee
SUBMITTED	Assign	UNDER_REVIEW	System
UNDER_REVIEW	Request Info	ADDITIONAL_INFO_REQUESTED	Reviewer

ADDITIONAL_INFO_REQUESTED	Respond	RESUBMITTED	Employee
RESUBMITTED	Resume Review	UNDER_REVIEW	System
UNDER_REVIEW	Approve	APPROVED	Approver
UNDER_REVIEW	Reject	REJECTED	Approver
APPROVED	Archive	ARCHIVED	System

5. Core Data Model

Entity	Primary Key	Important Fields	Relationships
User	user_id	name,email,role	Office,Department
Document	document_id	number,status,type	User
Attachment	attachment_id	file_name,path	Document
WorkflowInstance	workflow_id	status,current_stage	Document
WorkflowTask	task_id	assignee,due_date	WorkflowInstance
Comment	comment_id	type,text	Document
AuditLog	audit_id	action,timestamp	Document,User
Notification	notification_id	channel,status	User

6. Business Rules

BR-001: Document number must be unique.

BR-002: Only document owner may submit a draft.

BR-003: Approved documents cannot be modified.

BR-004: Audit records are immutable.

BR-005: Workflow stage SLA must be tracked.

BR-006: Additional document requests pause SLA timer.

BR-007: Only configured approvers can approve.

BR-008: Internal notes are not visible to request owner.

BR-009: Rejected documents cannot re-enter workflow without resubmission.

BR-010: Every state transition must generate an audit event.

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